Statistic 403/603

Mid-term exam 2

NAME:

STUDENT NUMBER:

SIGNATURE:

INSTRUCTIONS:

1. Do not panic. It rarely helps.
2. Read all questions carefully.
3. Calculators are permitted (none that have communication ability).
4. Where computations are required, you should show work.
5. If a question calls for an explanation, provide one that is complete.
6. Cross out any material you do not wish to have considered.
7. **Correct answers with insufficient justification or accompanied by additional incorrect statements will not receive full credit.**
8. Correct/incorrect answers without clearly displayed work cannot be considered for partial credit.
9. You may lose marks if your explanations are incomplete, missing or poorly worded.

In accordance with the SFU Academic Honesty Policy (T 10.02), academic dishonesty in any form will not be tolerated. Prohibited acts include, but are not limited to, the following:

- making use of any books, papers or electronic devices that allow communication,
- speaking or communicating with other students who are writing exams,
- copying the work of other students or purposely exposing written papers to the view of other students

Formulas:

Simple random sampling (SRS)

Sample mean: $\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n}$

Sample standard deviation: $s = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1}}$

Standard error of the mean: $se(\bar{y}) = \sqrt{(1 - \frac{n}{N})\frac{S^2}{n}}$

Estimate of the population total: $\hat{\tau} = N\bar{y}$

Standard error for the estimate of the population total: $se(\hat{\tau}) = N\sqrt{(1 - \frac{n}{N})\frac{S^2}{n}}$

Standard error for the estimate of the population proportion: $se(\hat{p}) = \sqrt{(1 - \frac{n}{N})\frac{P(1-P)}{n-1}}$

Sample size for SRS: $n \geq \frac{z_{\alpha/2}^2 s^2}{e^2 + \frac{z_{\alpha/2}^2 s^2}{N}}$

Stratified random sampling:

Estimate of the population mean (stratified random sampling): $\hat{\mu}_{str} = \bar{y}_{str} = \sum_{h=1}^{H} \frac{N_h}{N}\bar{y}_h = \sum_{h=1}^{H} W_h \bar{y}_h$

Standard error of stratified random sampling estimate of the pop. mean: $se(\mu_{str}) = \sqrt{\sum_{h=1}^{H} W_h^2 \frac{S_h^2}{n_h}(1 - n_h/N_h)}$

Cluster sampling:

Cluster sampling estimate of population mean: $\hat{\mu} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}$

Standard error of the cluster sampling estimator of the pop. mean: $se(\hat{\mu}) = \sqrt{\frac{1}{\bar{M}^2}\frac{S_{dif}^2}{n}(1 - n/N)}$

Cluster sampling estimator of pop. total: $\hat{\tau} = \hat{\mu}M$

Cluster sampling standard error for the estimator of the population total: $se(\hat{\tau}) = se(\hat{\mu})M$

1. (6 marks) A professor in the Biology Department is conducting a study on species diversity and stability. The professor constructs 9 representative systems that are similar to the larger environmental system in the real-world (these are called *microcosms*). To each microcosm, the professor randomly assigns three of each to a low, medium, and high diversity treatment (depending upon the number of plant and animal species included). These are grown in 3 growth chambers at constant temperatures for a year, and the treatments are randomly assigned within each chamber. The productivity and species richness at the end of the experiment are measured. The resulting arrangement of microcosms in the chambers was as follows:

Chamber 1	**Chamber 2**	**Chamber 3**
H	M	H
L	L	M
M	H	L

a. The professor analyzed the data using a one-way ANOVA testing for differences between the treatments. Explain why is this a problem?

If there is an effect of the chambers, it will not be taken into account. It may result in higher variability and increased chance of a type II error.

b. Write out the appropriate statistical model for this experiment, listing the appropriate assumptions.

(In this experiment, the Chamber is a block. The design is a randomized complete block design because each of the treatments appear in each)

$$y_{ij} = \mu_i + block_j + \epsilon_{ij}$$

i=1, 2, 3

j=1, 2, 3

$\epsilon_{ij} \sim N(0, \sigma)$

The errors are iid

2. The following quote was found in a scientific paper: "This experiment had 83% power to detect a difference in 3 kg in the mean growth among the treatments. The observed difference in the means was only 0.5 kg. Consequently, I am confident the null hypothesis was true."

a. Explain the first statement about power in terms that a non-statistician can understand.

Power: If the identical experiment were repeated a large number of times and if the true difference in the means was 3 kg, then in about 85% of the experiment a statistically significant result would have occurred and the null hypothesis of no difference would be rejected.

b. Do you agree with the last sentence? Why or why not?

The last sentence is not correct - the null hypothesis is rarely true (do you really think that the mean growth is identical to 10,000 decimal places?). Rather, these results would imply that the person was fairly confident that the difference in means was less than 3 kg.

3. According to Wikipedia, *Turbidity* is the cloudiness or haziness of a liquid caused by small particles that are generally invisible to the naked eye. The measurement of **turbidity** is a key test of water quality.

   Consider an experiment that aims to measure the impact of clear cutting forests upon the *turbidity* of streams and possible consequences on fish health. If the turbidity caused by clear cutting forests impacts fish health, the Department of Oceans and Fisheries will impose stricter environmental standards. On the other hand, if no impact on fish heath is found, nothing will be done.

   Steidhl et al. (1997) recommends against the uncritical use of hypothesis testing. In particular, he was disturbed by the failure to recognize the costs of Type I and Type II errors.

   Explain in non- technical language what these errors are (Type I and Type II errors), and the consequences of these errors in the context of experiments to measure the impact of clear cutting upon turbidity of streams and possible consequences on fish health.

   Solution: A Type I error is a false positive, i.e. we declare an effect exists when, in fact, no effect exists. In context, this would be to conclude that clearcutting has an effect upon stream turbidity, when in fact it has no effect. The consequences may be more stringent environmental protection laws than needed.

   A Type II error is a false negative, i.e. we fail to declare an effect exists when, in fact, an effect really does exist. In context, this would be to conclude that clearcutting has no effect upon stream turbidity, when in fact, it has an effect. The consequences may be that the increased turbidity negatively affects fish health leading to unacceptable environmental damage.

4. The yields of 2 varieties of corn (C1 and C2) and 2 fertilizers (F1 and F2) are to be compared using a greenhouse study. Within the greenhouse, there are 32 tables available for use arranged in a rectangular pattern of 8 by 4 tables. Eight replications of each treatment are to be done. The treatments are to be assigned to the tables where the corn is grown. After growth, the yields of each table are recorded (in kilograms).

   a. Show an experimental layout that uses a completely randomized design to investigate the effect of the two factors. What are the main advantages and disadvantages of this design?

   The advantage is that this is the easiest to conduct and requires the least complicated analysis.

C1F2	C2F1	C1F2	C2F2
C2F2	C1F2	C2F1	C1F1
C1F2	C1F2	C2F2	C1F1
C2F2	C2F2	C2F2	C2F1
C1F1	C1F1	C1F1	C2F2
C2F1	C2F1	C2F1	C2F1
C1F1	C2F1	C2F2	C1F2
C1F2	C1F1	C1F2	C1F1

   b. There is heating vent near one of the short sides of the rectangular pattern of 8 by 4 tables. The scientists feel that there may be an impact on the yield of being close to the vent. Show an experimental layout that uses a randomized block design to investigate the effect of the two factors. What are the main advantages of this design in comparison to the design in part a?

   Main advantage is that the impact of the vent is mitigated by the block. Each treatment is run at similar distances from the vent. This will increase the power if the vent has an impact on the response.

C1F1	C2F1	C1F2	C2F2
C1F1	C1F2	C2F1	C2F2
C2F1	C1F2	C2F2	C1F1
C2F2	C1F2	C1F1	C2F1
C1F1	C2F1	C2F2	C1F2
C2F1	C2F2	C1F2	C1F1
C1F1	C2F1	C2F2	C1F2
C1F2	C1F1	C2F2	C2F1

Vent

c. It turns out that, due to supply chain issues, the corn seed is only sold in small quantity bags (i.e., there is not a lot of seed in the bags). Fortunately, our resourceful scientists are able to get enough seed for their experiments by purchasing many smaller bags of corn seed from several suppliers. Unfortunately, each bag of seed contains only enough for two trials. To make matters worse, there is quite a bit of variation from bag to bag. Taking into account the air vent in part b., show an experimental layout that uses a split-plot design with main plots as a randomized complete block design. Write out the statistical model for this experiment.

In each row, red represents a bag of seed, and blue represents another bag. An entire bag of two seeds is used per row. So the red bag for the first row is not the same as any other row. We are still able to randomize the treatments among the four columns. One can view the seeds as a whole plot factor and the fertilizer as a subplot factor.

Here the blocks are still the rows which are equidistant from the vent.

$$y = \mu_i + \beta_j + \gamma_{ij} + block_k + \epsilon_l + e_{ijkl}$$

i = 1,2 (seed variety)
j = 1,2 (fertilizer variety)
k = 1, 2, …8 (rows)
l = 1, 2, …16 (bags)
$\epsilon_l \sim N(0, \sigma_{whole-plot})$
$e_{i,j,k,l} \sim N(0, \sigma_{sub-plot})$
The errors are all independent

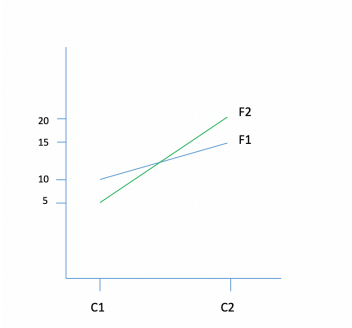
C2F1	C1F2	C2F2	C1F1
C1F2	C1F1	C2F1	C2F2
C2F1	C2F2	C1F2	C1F1
C2F1	C1F1	C2F2	C1F2
C2F2	C2F1	C1F1	C1F2
C2F1	C1F2	C2F2	C1F1
C1F1	C1F2	C2F2	C2F1
C1F2	C2F1	C2F2	C1F1

Vent

d. Suppose that the experiment in part a. was performed (ignore parts b. and c.). The following table shows the mean responses for each of the four treatments.

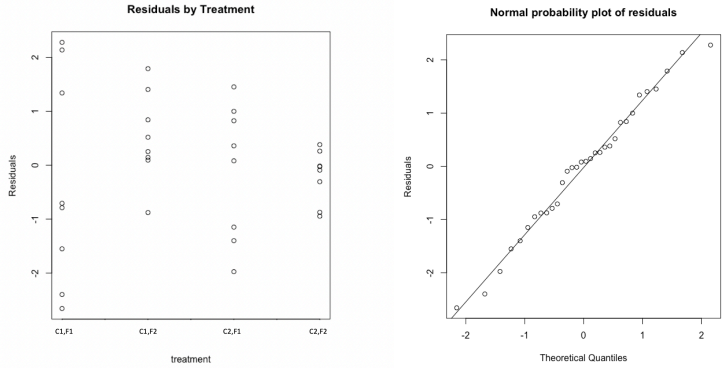
		Corn variety	
		C1	**C2**
Fertilizer type	**F1**	10	15
	F2	20	5

Draw an interaction plot (or profile plot) for these results (label all axes and lines). Comment on whether or not you would expect to see a significant interaction effect.



Since the lines are not parallel, one would anticipate a significant interaction effect.

e. Following the experiment, the scientists fit a 2-factor ANOVA model for the experiment. Before continuing with their analysis, they decided to investigate the residuals. Below are some plots of the residuals. Comment on whether or not any of the model assumptions appear to be violated.



It appears that the normality assumption holds. The *identically distributed* assumption appears violated since the variability in the C2F2 observations appears smaller than the other groups